The Blackwell Information-Metric Unification (ODIM-U v1.2) Observer-Driven Informational Emergence of Spacetime from a Relative Entropy Action

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Abstract

This paper proposes the Blackwell Information-Metric Unification (ODIM-U v1.2) framework as a conceptual synthesis inspired by entropic gravity approaches (e.g., Verlinde 2010; Jacobson 1995) and recent quantum information-based models of gravity (e.g., Bianconi 2025). The model explores gravitational dynamics and the emergence of time (Δt) through minimization of quantum relative entropy S_{rel} between a quantum field state (ρ) and an effective classical reference state (σ).

This process is governed by a Symmetry Information Constraint ($\mathcal{C}_{\text{sym}}$) enforcing monotonic decrease of informational distinguishability, yielding a local gravitational potential ($\Psi < 0$). Time differentials emerge from decoherence rates (Γ) tied to gravitational effects, recovering standard gravitational time dilation in weak-field limits. The framework proposes a modified Einstein equation with an informational stress-energy term and suggests testable predictions via precision experiments.

Keywords: Entropic Gravity, Quantum Information, Relative Entropy, Emergent Spacetime, Decoherence, Quantum Gravity.

1 Theoretical Foundation: The Relative Entropy Action

1.1 The Informational Core: Relative Entropy

Building on quantum relative entropy as a measure of informational distinguishability (central to recent entropic gravity models, e.g., Bianconi 2025), the gravitational potential (Ψ) arises from local mismatches between the quantum density matrix (ρ) and a classical reference (σ):

$$S_{\mathrm{rel}}(\rho||\sigma) = \mathrm{Tr}[\rho(\ln\rho - \ln\sigma)] \tag{1}$$

The Symmetry Information Constraint enforces monotonic decrease of relative entropy in dynamical evolution, consistent with its fundamental property under completely positive trace-preserving (CPTP) maps in quantum channels:

$$\mathcal{C}_{\mathrm{sym}} : \frac{dS_{\mathrm{rel}}}{dt} \leq 0 \tag{2}$$

This constraint drives the system toward reduced informational mismatch, with Ψ emerging as the effective (negative) potential encoding this process. For illustration, in weak fields, σ approximates the flat Minkowski metric, while deviations in ρ induced by matter generate curvature-like effects.

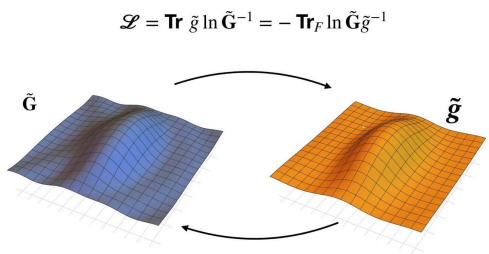

Figure 1: Enter Caption

Figure 2: Diagrammatic representation of the entropic quantum gravity action as quantum relative entropy between the spacetime metric and the matter-induced metric (inspired by Bianconi 2025).

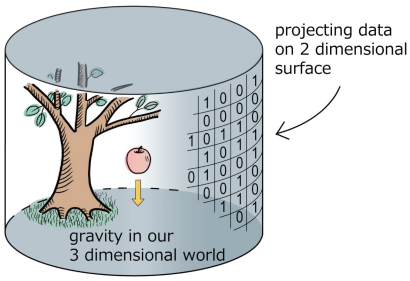

Figure 3: Enter Caption

Figure 4: Conceptual illustration of emergent gravity from quantum informational mismatch.

1.2 Emergence of Time (Δt)

Observed time differentials emerge from the decoherence rate (Γ), with stronger (more negative) gravitational potentials leading to slower proper time (standard dilation). Inspired by open quantum systems in curved spacetime and entropic models:

$$\Gamma \propto \frac{k_B T_{\text{eff}}}{\hbar} \left(1 + c_1 \frac{\Psi}{c^2} + \cdots\right) \tag{3}$$

(where $\Psi < 0$, and $c_1 > 0$ is a positive constant). In the weak-field limit, this yields:

$$\Delta t_{\text{obs}} \propto \frac{1}{\Gamma} \propto 1 - \frac{\Psi}{c^2} + \cdots \approx \sqrt{1 + 2\Psi/c^2} \tag{4}$$

recovering the general relativistic time dilation factor (with T_{eff} as an effective Unruh-like temperature or environmental bath). This represents the effective duration for quantum-to-classical transitions, linking informational processing to proper time.

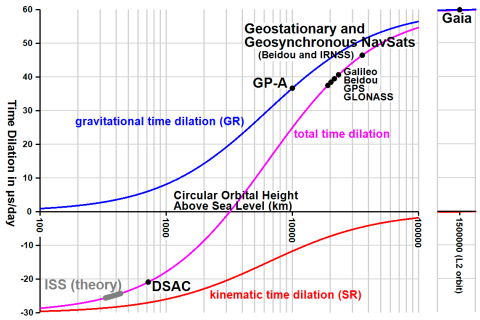

Figure 5: Enter Caption

Figure 6: Enter Caption

Figure 7: Gravitational time dilation: clocks at lower altitude run slower than those at higher altitude.

2 The Information-Metric Field Equation (IMFE)

The metric dynamics follow a modified Einstein equation:

$$\mathbf{G}_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}(\mathbf{T}_{\mu\nu} + \mathbf{T}_{\text{info}}) \tag{5}$$

The informational tensor sources gravity from relative entropy gradients (similar to scalar-field-like terms in entropic models; recovers standard GR in weak-information limits):

$$\mathbf{T}_{\text{info}} \propto (\nabla_\mu \mathcal{S}_{\text{rel}})(\nabla_\nu \mathcal{S}_{\text{rel}}) \tag{6}$$

This quadratic form resembles kinetic terms for effective fields and generally satisfies weak/null energy conditions in low-curvature regimes (as gradients are small), though strong-field violations could emerge as quantum corrections—worthy of further study.

This aligns with derivations of gravity from quantum relative entropy (e.g., Bianconi 2025) while incorporating observer-driven contextualization.

3 The Role of Context and Measurement

Observer effects are modeled via a contextual efficiency factor ($\beta_{\text{meas}} \in [0, 1]$), representing apparatus coherence and measurement precision (objective and operational, e.g., via noise levels or entanglement fidelity):

$$S_{\text{rel}}^{\text{eff}}(\rho||\sigma) = \frac{1}{\beta_{\text{meas}}} S_{\text{rel}}^{\text{ideal}} \tag{7}$$

High $\beta_{\text{meas}} \to 1$ (coherent, low-noise setups) minimizes effective entropy, approaching ideal dynamics. This is covariant, aligning with relational quantum mechanics interpretations without subjectivity or preferred frames. Future work should specify operational protocols for β_{meas} (e.g., via decoherence functionals).

4 Falsifiable Predictions

The framework is testable beyond standard GR/QM:

Prediction 1: Gravitationally Modified Decoherence Deviations in decoherence rates for entangled systems in gravitational fields (e.g., underground/satellite optomechanics), with distinguishing scaling beyond pure time-dilation redshift. The proposed weak-field expansion could yield measurable shifts in interference visibility, constrained by current bounds but probeable with next-generation experiments.

Prediction 2: Propagation Anomalies in Low-Density Media Subtle timing anomalies in pulsar/FRB signals correlated with interstellar gradients, potentially exceeding known plasma dispersion, scattering, or lensing if informational in origin (orthogonal to standard effects):

$$\Delta t_{\text{anomaly}} \propto \int_{\text{path}} \Gamma^{-1}\, dl \tag{8}$$

Quantitative modeling against archival data (e.g., CHIME/FAST) could falsify or support.

5 Conclusion: Toward a Unified Informational Framework

ODIM-U v1.2 synthesizes entropic and quantum-information approaches to gravity, providing an emergent model compatible with general relativity in weak limits. It draws connections to ongoing work (e.g., Bianconi's entropic action and G-field) while exploring observer-contextual decoherence. Further iterations should derive the Γ-Ψ scaling rigorously (e.g., via modular flows or open-system master equations) and analyze $\mathbf{T}_{\text{info}}$ energy conditions in detail. The framework invites empirical scrutiny through precision timing, interference, and cosmological observations.

References

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